

Supplementary Materials to “Integration of Vital Signs Monitoring System Based on Single Antenna Bio-Sensing (SABioS) Method”

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A. CALCULATING FIELD STRENGTH ON A SPECIFIC DISTANCE BASED ON EIRP

Having the Effective Isotropic Radiated Power (EIRP) of the antenna, the power density at the distance r can be written as:

$$P_d(r) = \frac{EIRP}{4\pi r^2}$$

The $P_d(r)$ can also be expressed as [1]:

$$P_d(r) = \frac{EIRP}{4\pi r^2} = \frac{E^2}{120\pi}$$

where E is electric field, and 120π is the impedance of free space. Therefore $|E|$ at r meters away from the radiator is:

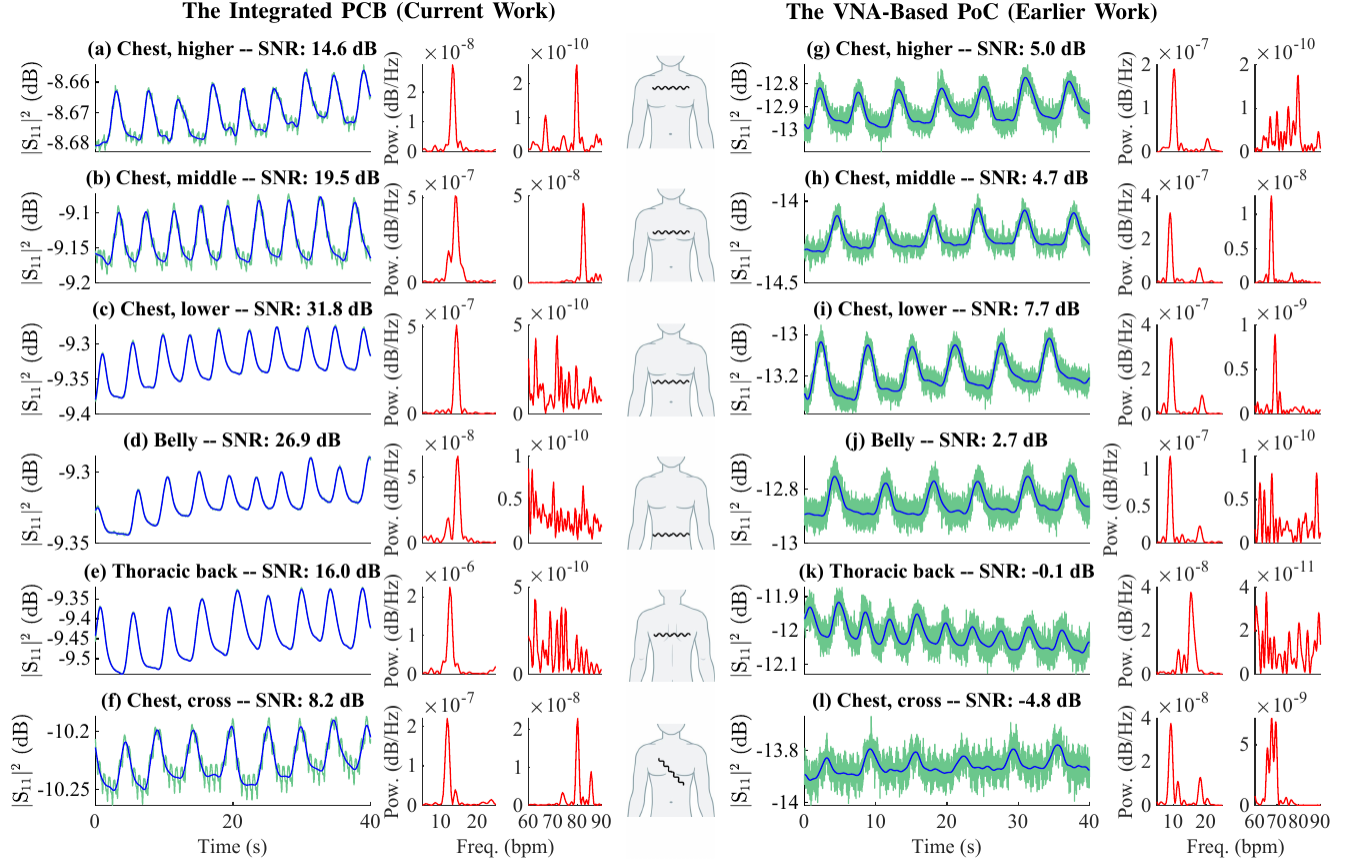
$$|E| = \sqrt{30 \frac{EIRP}{r^2}}$$

Now, plugging in $r = 3 \text{ m}$,

$$|E|_{3\text{m}} = \sqrt{\frac{10}{3} EIRP}.$$

Therefore, for an $EIRP$ of -20.49 dBm equalling to 8.933×10^{-6} watts, field strength at 3 meters equals 5.45 mV/m .

B. MONITORING ON DIFFERENT ANTENNA LOCATIONS – CURRENT WORK VS EARLIER PROOF-OF-CONCEPT WORK



References

- [1] C. A. Balanis, *Antenna Theory: Analysis and Design*. Somerset: Wiley, 2012.